

request&json

May 10, 2026

```
[ ]: ## requests and json
import requests
import json
import pandas as pd
# Basic GET request
url = "https://jsonplaceholder.typicode.com/posts/1"
response = requests.get(url)
print(response)
# Check status code
print(response.status_code)
# View raw response text
print(response.text)
# Convert response to JSON
data = response.json()
print(data)
print(type(data))
# Access JSON values
print(data["userId"])
print(data["title"])
print(data["body"])
# Pretty print JSON
formatted_json = json.dumps(data, indent=4)
print(formatted_json)
# Get multiple posts
url = "https://jsonplaceholder.typicode.com/posts"
response = requests.get(url)
posts = response.json()
print(type(posts))
print(len(posts))
# Loop through JSON list
for post in posts[:5]:
    print(post["id"])
    print(post["title"])
    print("-" * 50)
# Send query parameters
params = {
    "userId": 1
```

```

}
response = requests.get(
    "https://jsonplaceholder.typicode.com/posts",
    params=params
)
print(response.url)
filtered_posts = response.json()
print(len(filtered_posts))
# Send custom headers
headers = {
    "User-Agent": "Mozilla/5.0",
    "Accept": "application/json"
}
response = requests.get(
    "https://jsonplaceholder.typicode.com/posts/1",
    headers=headers
)
print(response.status_code)
# Send POST request
new_post = {
    "title": "Learning Requests",
    "body": "This is tutorial content",
    "userId": 1
}
response = requests.post(
    "https://jsonplaceholder.typicode.com/posts",
    json=new_post
)
print(response.status_code)
print(response.json())
# Send PUT request
updated_post = {
    "id": 1,
    "title": "Updated Title",
    "body": "Updated body",
    "userId": 1
}
response = requests.put(
    "https://jsonplaceholder.typicode.com/posts/1",
    json=updated_post
)
print(response.json())
# Send PATCH request
patch_data = {
    "title": "Patched Title"
}
response = requests.patch(

```

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        "https://jsonplaceholder.typicode.com/posts/1",
        json=patch_data
    )
    print(response.json())
    # Send DELETE request
    response = requests.delete(
        "https://jsonplaceholder.typicode.com/posts/1"
    )
    print(response.status_code)
    # View response headers
    response = requests.get(
        "https://jsonplaceholder.typicode.com/posts/1"
    )
    print(response.headers)
    # Check content type
    print(response.headers["Content-Type"])
    # Download image using binary content
    image_url = "https://picsum.photos/200"
    response = requests.get(image_url)
    with open("random_image.jpg", "wb") as f:
        f.write(response.content)
    print("Image downloaded")
    # Save JSON data to file
    response = requests.get(
        "https://jsonplaceholder.typicode.com/posts/1"
    )
    data = response.json()
    with open("post.json", "w") as f:
        json.dump(data, f, indent=4)
    print("JSON file saved")
    # Read JSON data from file
    with open("post.json", "r") as f:
        loaded_data = json.load(f)
    print(loaded_data)
    # Convert API response to DataFrame
    response = requests.get(
        "https://jsonplaceholder.typicode.com/posts"
    )
    posts = response.json()
    df = pd.DataFrame(posts)
    print(df.head())
    # Loop through API data
    for post in posts[:5]:
        print("ID:", post["id"])
        print("TITLE:", post["title"])
        print("BODY:", post["body"])
        print("=" * 50)

```

```

# Get nested JSON response
response = requests.get(
    "https://jsonplaceholder.typicode.com/users/1"
)
user = response.json()
print(user)
# Access nested JSON values
print(user["name"])
print(user["email"])
print(user["address"]["city"])
print(user["company"]["name"])
# Convert JSON string into Python object
json_string = '''
{
    "name": "Alice",
    "age": 25
}
'''
python_data = json.loads(json_string)
print(python_data)
print(type(python_data))
# Convert Python object into JSON string
person = {
    "name": "Bob",
    "age": 30
}
json_output = json.dumps(person, indent=4)
print(json_output)
# Show common HTTP status codes
print("200 = OK")
print("201 = Created")
print("400 = Bad Request")
print("401 = Unauthorized")
print("403 = Forbidden")
print("404 = Not Found")
print("500 = Server Error")

```

API FUNDA

May 20, 2026

```
[ ]: # 3. Tokens (Authorization)
import requests
url = "https://jsonplaceholder.typicode.com/posts"
token = "my_secret_token"
headers = {
    "Authorization": f"Bearer {token}"
}
response = requests.get(url, headers=headers)
print(response.status_code)
```

```
[ ]: # Pagination
import requests
url = "https://jsonplaceholder.typicode.com/posts"
params = {
    "_page": 1,
    "_limit": 5
}
response = requests.get(url, params=params)
data = response.json()
print("Posts on page 1")
for post in data:
    print(post["id"], post["title"])
```

```
[ ]: ## Multiple pages
import requests
base_url = "https://jsonplaceholder.typicode.com/posts"
page = 1
while page <= 3:
    params = {
        "_page": page,
        "_limit": 5
    }
    response = requests.get(base_url, params=params)
    posts = response.json()
    print(f"\nPage {page}")
    for post in posts:
        print(post["id"], post["title"])
    page += 1
```

APIFUNDA-Copy1

May 25, 2026

```
[ ]: import requests
r = requests.get("https://api.github.com/rate_limit")
data = r.json()
print("Limit :", data["rate"]["limit"])
print("Used  :", data["rate"]["used"])
print("Left   :", data["rate"]["remaining"])
print("Reset  :", data["rate"]["reset"])
```

```
[ ]: import requests
for i in range(70):

    r = requests.get("https://api.github.com/users/octocat")

    print(i + 1, r.status_code)

    # GitHub rate limit exceeded
    if r.status_code == 403:
        data = r.json()

        if "API rate limit exceeded" in data.get("message", ""):
            print("Rate limit exceeded!")
            print(data["message"])
            break
```

djangomodels

May 29, 2026

```
structure
vehicleapp/

    models.py
    views.py
    urls.py

    templates/
        vehicleapp/
            base.html
            home.html
            vehicle_detail.html

    admin.py

project - vehiclehub
app - vehicleapp

- Register app
## vehicleapp models.py
from django.db import models
from django.contrib.auth.models import User

# Create your models here.
from django.db import models
from django.contrib.auth.models import User

class Feature(models.Model):
    name = models.CharField(max_length=50)

    def __str__(self):
        return self.name

class Vehicle(models.Model):
    FUEL_CHOICES = [
        ('petrol', 'Petrol'),
        ('diesel', 'Diesel'),
        ('electric', 'Electric')
    ]
```

```

owner = models.ForeignKey(User, on_delete=models.CASCADE)
features = models.ManyToManyField(Feature, blank=True)

make = models.CharField(max_length=50)
model_name = models.CharField(max_length=50)
year = models.PositiveIntegerField()
price = models.DecimalField(max_digits=10, decimal_places=2)

fuel_type = models.CharField(
    max_length=20,
    choices=FUEL_CHOICES
)

is_sold = models.BooleanField(default=False)

description = models.TextField(
    blank=True,
    null=True
)

created_at = models.DateTimeField(
    auto_now_add=True
)

def __str__(self):
    return f"{self.make} {self.model_name}"

class Meta:
    ordering = ['-created_at']

## view.py
from django.shortcuts import render, get_object_or_404
from .models import Vehicle

def home(request):
    vehicles = Vehicle.objects.all()

    context = {
        "vehicles": vehicles,
        "total": Vehicle.objects.count()
    }

    return render(
        request,
        "vehicleapp/home.html",
        context
    )

```



```

def vehicle_detail(request, vehicle_id):
    vehicle = get_object_or_404(
        Vehicle,
        id=vehicle_id
    )

    return render(
        request,
        "vehicleapp/vehicle_detail.html",
        {"vehicle": vehicle}
    )

## project urls
from django.contrib import admin
from django.urls import path, include
urlpatterns = [path("admin/", admin.site.urls), path("", include("vehicleapp.urls"))]

## vehicleapp.urls
from django.urls import path
from . import views

urlpatterns = [
    path('', views.home, name='home'),
    path(
        'vehicle/<int:vehicle_id>/',
        views.vehicle_detail,
        name='vehicle_detail'
    ),
]

## base.html
{% load static %}
<!DOCTYPE html>
<html>
<head>
<title>VehicleHub</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
    rel="stylesheet">
</head>
<body>

{% include 'vehicleapp/navbar.html' %}

<div class="container mt-4">
{% block content %}
{% endblock %}

```

```

</div>

</body>
</html>

## navbar.html
<nav class="navbar navbar-dark bg-dark">
<div class="container">
<a class="navbar-brand"
    href="{% url 'home' %}">
    VehicleHub
</a>
</div>
</nav>

## home.html
{% extends 'vehicleapp/base.html' %}

{% block content %}

<h2>Total Vehicles: {{ total }}</h2>

<table class="table table-striped">

<tr>
<th>Vehicle</th>
<th>Fuel</th>
<th>Status</th>
<th></th>
</tr>

{% for vehicle in vehicles %}

<tr>

<td>
{{ vehicle.make }}
{{ vehicle.model_name }}
</td>

<td>
{{ vehicle.fuel_type }}
</td>

<td>

{% if vehicle.is_sold %}
Sold

```

```

{% else %}
Available
{% endif %}

</td>

<td>
<a href="{% url 'vehicle_detail' vehicle.id %}"
    class="btn btn-primary">
View
</a>
</td>

</tr>

{% endfor %}

</table>

{% endblock %}

## Features details
{% extends 'vehicleapp/base.html' %}

{% block content %}

<h2>
{{ vehicle.make }}
{{ vehicle.model_name }}
</h2>

<ul class="list-group">

<li class="list-group-item">
Owner: {{ vehicle.owner }}
</li>

<li class="list-group-item">
Price: {{ vehicle.price }}
</li>

<li class="list-group-item">
Fuel: {{ vehicle.fuel_type }}
</li>

<li class="list-group-item">
Description:
{{ vehicle.description }}

```

```

</li>

<li class="list-group-item">

Features:

{% for feature in vehicle.features.all %}

<span class="badge bg-success">
{{ feature.name }}
</span>

{% empty %}

No Features

{% endfor %}

</li>

</ul>

{% endblock %}

## admin
from django.contrib import admin
from .models import Vehicle, Feature
admin.site.register(Vehicle)
admin.site.register(Feature)

## Createsuperuser
python manage.py createsuperuser

## shell commands
# Start shell:
# python manage.py shell

from django.contrib.auth.models import User
from vehicleapp.models import Vehicle, Feature

# Create User
user = User.objects.create_user(
    username='manish',
    password='123456'
)

# Create Features
gps = Feature.objects.create(name='GPS')

```

```
sunroof = Feature.objects.create(name='Sunroof')
airbags = Feature.objects.create(name='Airbags')
```

```
# Create Vehicles
```

```
v1 = Vehicle.objects.create(
    owner=user,
    make='Tesla',
    model_name='Model S',
    year=2024,
    price=9000000,
    fuel_type='electric',
    is_sold=False,
    description='Premium electric sedan'
)
```

```
v2 = Vehicle.objects.create(
    owner=user,
    make='Honda',
    model_name='City',
    year=2022,
    price=1500000,
    fuel_type='petrol',
    is_sold=True,
    description='Family sedan'
)
```

```
v3 = Vehicle.objects.create(
    owner=user,
    make='Tata',
    model_name='Nexon EV',
    year=2024,
    price=1800000,
    fuel_type='electric',
    is_sold=False,
    description='Electric SUV'
)
```

```
# Add Many-to-Many Features
```

```
v1.features.add(gps, sunroof, airbags)
v2.features.add(airbags)
v3.features.add(gps)
```

```
# ORM Queries
```

```
# Display all vehicles
Vehicle.objects.all()
```

```
# Count records
```

```

Vehicle.objects.count()

# Filter electric vehicles
Vehicle.objects.filter(
    fuel_type='electric'
)

# Filter available (unsold) vehicles
Vehicle.objects.filter(
    is_sold=False
)

# Order by price ascending
Vehicle.objects.order_by(
    'price'
)

# Order by price descending
Vehicle.objects.order_by(
    '-price'
)

# Get vehicle with ID 1
Vehicle.objects.get(
    id=1
)

# Access ForeignKey
v1.owner

# Access Many-to-Many relationship
v1.features.all()

# Reverse lookup
gps.vehicle_set.all()

# Update a record
v1.price = 8500000
v1.save()

# Verify update
Vehicle.objects.get(id=1)

# Delete a record
v2.delete()

# Verify delete
Vehicle.objects.all()

```

```
# Exit shell  
exit()
```

[]:

Djangotemplates

May 29, 2026

```
## project structure
```

```
vehicleapp/
```

```
    templates/
```

```
        vehicleapp/
```

```
            base.html
```

```
            home.html
```

```
            contact.html
```

```
            navbar.html
```

```
    static/
```

```
        vehicleapp/
```

```
            css/
```

```
                style.css
```

```
python -m django startproject vehicleproject
```

```
cd vehicleproject
```

```
python manage.py runserver
```

```
python manage.py startapp vehicleapp
```

```
- Register app
```

```
## vehicleapp/views.py
```

```
from django.shortcuts import render
```

```
def home(request):
```

```
    vehicles = [
```

```
        {"id": 1, "name": "Tesla Model S", "price": 9000000, "premium": True},
```

```
        {"id": 2, "name": "Honda City", "price": 1500000, "premium": False},
```

```
        {"id": 3, "name": "BMW X5", "price": 9500000, "premium": True},
```

```
    ]
```

```
    context = {
```

```
        "title": "Vehicle List",
```

```
        "vehicles": vehicles,
```

```
        "user_name": "Manish"
```

```
    }
```

```
    return render(request, "vehicleapp/home.html", context)
```

```
def contact(request):
```



```

    message = ""
    if request.method == "POST":
        message = request.POST.get("name")
    return render(
        request,
        "vehicleapp/contact.html",
        {"message": message}
    )
## Vehicleapp urls
from django.urls import path
from . import views
urlpatterns = [
    path('', views.home, name='home'),
    path('contact/', views.contact, name='contact'),
]

## templates/vehicleapp/base.html
{% load static %}
<!DOCTYPE html>
<html>
<head>
    <title>{{ title }}</title>
    <link rel="stylesheet"
        href="{% static 'vehicleapp/css/style.css' %}">
</head>
<body>
    {% include 'vehicleapp/navbar.html' %}
    <hr>
    {% block content %}
    {% endblock %}
</body>
</html>

## templates/vehicleapp/navbar.html
<a href="{% url 'home' %}">Home</a> |
<a href="{% url 'contact' %}">Contact</a>
<hr>

## templates/vehicleapp/home.html
{% extends 'vehicleapp/base.html' %}
{% block content %}
<h1>Welcome {{ user_name }}</h1>
<h2>{{ title }}</h2>
<ul>
{% for vehicle in vehicles %}
    <li>
        {{ vehicle.name }}
        - {{ vehicle.price }}

```

```

        {% if vehicle.premium %}
            Premium Vehicle
        {% endif %}
    </li>
{% endfor %}
</ul>
{% endblock %}

## templates/vehicleapp/contact.html
{% extends 'vehicleapp/base.html' %}
{% block content %}
<h2>Contact Form</h2>
<form method="POST">
    {% csrf_token %}
    <label>Name:</label>
    <input type="text"
        name="name">
    <button type="submit">
        Submit
    </button>
</form>
{% if message %}
    <h3>Hello {{ message }}</h3>
{% endif %}
{% endblock %}

## static/vehicleapp/css/style.css
body{
    font-family: Arial;
    margin: 20px;
}

h1{
    color: blue;
}

li{
    margin: 10px;
}

## BOOTSTRAP
vehicleapp/

static/
    vehicleapp/
        css/
            bootstrap.min.css
            style.css

```

```

        js/
        bootstrap.bundle.min.js
## ADD IN BASE HTML
{% load static %}

<link rel="stylesheet"
      href="{% static 'vehicleapp/css/bootstrap.min.css' %}">

<link rel="stylesheet"
      href="{% static 'vehicleapp/css/style.css' %}">
AND AT BOTTOM OF BASE HTML

<script src="{% static 'vehicleapp/js/bootstrap.bundle.min.js' %}"></script>

FOR BOOTSTRAP STYLED SUBMIT BUTTON ADD -> class="btn btn-primary"

```

[]:

pb442

May 26, 2026

0.1 PB 442

```
django-admin startproject myproject
cd myproject
python manage.py roll
```

```
## roll/models.py
from django.db import models
class Student(models.Model):
    name = models.CharField(max_length=100)
    email = models.EmailField()
    result = models.CharField(max_length=50)
    def __str__(self):
        return self.name
```

```
##migrations
python manage.py makemigrations
python manage.py migrate
```

```
# roll/admin.py
from django.contrib import admin
from .models import Student
```

```
admin.site.register(Student)
```

```
## superuser
python manage.py createsuperuser
```

```
## roll/views.py
from django.shortcuts import render, get_object_or_404
from .models import Student
# 1. student_search
def student_search(request):
    query = request.GET.get('q')
    students = Student.objects.all()
    if query:
        students = students.filter(name__icontains=query)
```

```

        context = {
            'students': students,
            'query': query
        }
        return render(request, 'roll/home.html', context)
# 2. Studentinfo
def Studentinfo(request):
    students = Student.objects.all()
    return render(request, 'roll/home.html', {'students': students})
# 3. welcome
def welcome(request):
    return render(request, 'roll/welcome.html')
# 4. stuinformation
def stuinformation(request, id):
    student = get_object_or_404(Student, id=id)
    return render(request, 'roll/information.html', {'student': student})

## roll/urls.py
from django.urls import path
from . import views
urlpatterns = [
    path('', views.Studentinfo, name='home'),
    path('search/', views.student_search, name='student_search'),
    path('welcome/', views.welcome, name='welcome'),
    path('information/<int:id>/', views.stuinformation, name='stuinformation'),
]

## myproject urls
from django.contrib import admin
from django.urls import path, include
urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('roll.urls')),
]

## templates/roll/home.html
<!DOCTYPE html>
<html>
<head>
    <title>Home</title>
</head>
<body>
    <!-- NAVBAR -->
    <a href="/">Home</a>
    <a href="/welcome/">welcome</a>
    <h1>my roll number (here write your roll number)</h1>
    <h1>Student Page</h1>
    <!-- SEARCH FORM -->

```

```

<form action="/search/" method="GET">
    <input type="text" name="q" placeholder="Enter name">
    <input type="submit" value="search">
</form>
<br>
<h1>Show Data</h1>
<table border="1" cellpadding="10">
    <tr>
        <th>ID</th>
        <th>StuID</th>
        <th>Name</th>
        <th>Email</th>
        <th>Result</th>
        <th>Information</th>
    </tr>
    {% if students %}
        {% for student in students %}
            <tr>
                <td>{{ student.id }}</td>
                <td>{{ student.name }}</td>
                <td>{{ student.email }}</td>
                <td>{{ student.result }}</td>
                <td>
                    <a href="/information/{{ student.id }}/">
                        <button>information</button>
                    </a>
                </td>
            </tr>
        {% endfor %}
    {% else %}
        <tr>
            <td colspan="6">
                NO DATA
            </td>
        </tr>
    {% endif %}
</table>
</body>
</html>

## welcome.html
<!DOCTYPE html>
<html>
<head>
    <title>WELCOME</title>
</head>
<body>
    <a href="/">Home</a>

```

```
<a href="/welcome/">welcome</a>
<h1>my roll number (here write your roll number)</h1>
<h1>I AM STUDENT</h1>
</body>
</html>
```

information

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Information</title>
</head>
<body>
  <a href="/">Home</a>
  <a href="/welcome/">welcome</a>
  <h1>Student Information</h1>
  <h2>my roll number is {{student.id}}</h2>
  <h2>My name is {{student.name}}</h2>
  <h2>My email is {{student.email}}</h2>
</body>
</html>
```

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